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Enhancement of Charging Efficiency of Batteries for Electric Vehicles: Review

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Abstract— **To minimize emissions of greenhouse gases, the electric vehicle industry has grown dramatically in recent years. Traditional vehicles are improved by a variety of technologies, including regenerative braking, auto-start and shut-off, and even more advanced technology. Affordable charging stations with state-of-the-art control algorithms are required to boost the charging efficiency of electric vehicles (EVs). An electronic regulator known as a "Battery Management System" (BMS) controls battery monitoring and protection, calculation of battery status, and potential user notification of that condition. Due to many faults in battery packs, EVs encounter a number of issues; it is essential to assess the battery's level of charge (SOC). Measurement of battery performance and health are part of battery cell monitoring. The ideas of artificial intelligence (AI) and machine learning (ML) for battery management in new energy vehicles were initially presented to address the issue of protecting battery cells from over-current and over-voltage. Data is collected by applying multiline signals to one-of-a-kind dynamic charge and discharge test. Machine learning (ML) algorithms are helpful in state of charge (SOC) prediction challenges because they can record cell dynamics and store past data, both of which are necessary for predicting future charge levels. Along with different charging methods, energy storage technologies, BMS topologies, multiple ML algorithms and Battery Management System (BMS) used in an electric car arealso explored in this review.**

Keywords:Charging efficiency, Electric vehicles, Battery management systems, Machine learning.

I. Introduction

A future of transportation systems and decreased carbon dioxide emission is being paved with the aid of electric vehicles (EVs). However, concerns about range and charging times are now limiting the usage of electric vehicles. Fast-charging electrical energy storage systems, which are mostly in the form of Li-Ion battery packs, are therefore a crucial enabling technology for EVs[1].Range anxiety has been identified as one of the most significant hurdles in electric vehicles during the previous decade [2]. Extending the battery's life and expanding its holding capacity to improvebattery efficiency by modeling the balancing architecture will show how quick charging affects the electrical, thermal, and degrading responses of batteries [3-5].Charging stations are categorized into classes depending on how much power they use, and different optimization tactics, approaches, and future perspectives are offered in order to obtain the best design feasible. Following that, the language for fast charging is employed[6]-[9]. There are already a variety of approaches for determining health-conscious rapid charging strategies. Selecting the strategy that is best for a given application requires assessing and contrasting each method's unique mix of advantages and disadvantages. From the perspective of a charging algorithm, fast charging and its application have been examined at the material level, as well as at the cell and system level [10]-[12].

This review deals with introduction first which gives a brief explanation of the goal of the paper. The overview of electric cars in section 2 covers their various types and effects. Electric vehicle charging is covered in section 3. Here, three distinct charge levels are shown. Section 4 of the review is dedicated to the functions of energy storage systems. It consists of several BMS topologies for faster electric vehicle charging.Section5 is dedicated to various ML approaches to BMS. The conclusion is presented in section 6.

II. OVERVIEW OF ELECTRIC VEHICLES

Two or more electric motors, powered by internal rechargeable batteries are used to drive an electric vehicle. An electric vehicle is a sensible choice for maintaining a pollution-free environment. It has been noted that moving to electric cars from internal combustion engines decreases net CO_2 emissions [1]. To increase vehicle range and allay concerns about the charging process, a suitable charging station must be offered to electric car makers [13].

A. Hybrid Electric Vehicles (HEV):

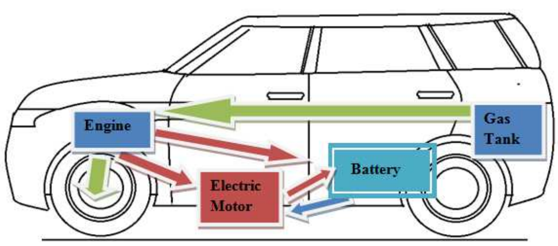

Fig. 1 HEV

Fig.1 shows an HEV, a vehicle that combines internal combustion engine technology with electric power technology. It can operate with a high-voltage battery and drive a vehicle with an electrical load of up to 80 kW. The amount of time an HEV can run on electricity rather than gasoline is used to calculate its fuel efficiency [14].

B. Plug-in Hybrid Electric Vehicles (PHEV):

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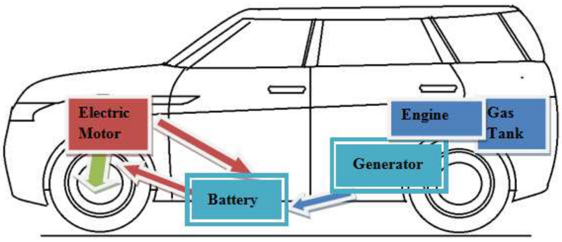

Fig. 2 PHEV

Regenerative braking technology is used in hybrid electric vehicles to recharge batteries, and when necessary, engine combustion produces electricity. A PHEV's battery may be charged from outside sources like a cable connection or an inductive pad, substantially lowering fuel usage as shown in Fig.2. The battery may be recharged often since PHEVs have such high fuel efficiency. After HEV, PHEV is the variant with the second-fastest growth [15].

C.Fuel Cell EV (FCEV):

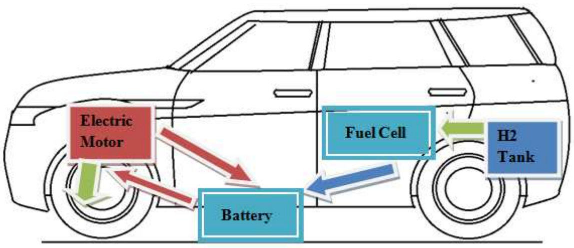

Fig. 3 FCEV

Both electric vehicles and FCEVs employ the same propulsion technology. In place of a battery, a fuel cell powers the onboard electric motor as shown in Fig.3. As hydrogen and oxygen are compressed, energy is produced by the fuel cell. However, hydrogen is toxic to store and transport, it is not pollution-free [16].

D. Battery Electric Vehicle (BEV):

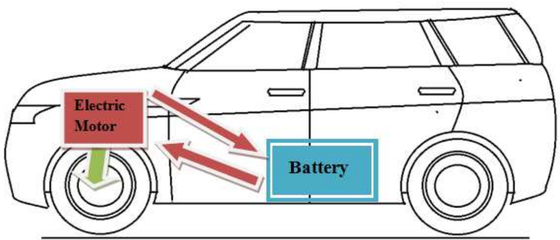

Fig. 4 BEV

"Battery-electric vehicle" utilizing the battery reduces fuel use to a minimum but reduces vehicle range as compared to alternative methods as shown in Fig.4. Battery electric vehicles will soon be readily accessible, and HEVs are currently simple to find [17].

An overview of four vehicle types is provided in Table 1.

TABLE 1. PARAMETERS OF ELECTRIC VEHICLES

Parameter	HEV	PHEV	FCEV	BEV
Power source	Primary gasoline	Gasoline, batteries	Hydrogen fuel	Batteries
Driving component	Electric motor, ICE	Electric motor, gasoline	Electric motor	Electric motor
Battery capacity in kWh	0.65-1.8	4.4-34	10-17	16-100
Grid connected	No	Yes	No	Yes
Electric range in miles	0-50	10-50	upto 366	upto 400
CO2 emission	0-50 lbs of CO/100 miles	0-50 lbs of CO/100 miles	No	No
External plug-in	No	Yes	No	Yes

III. CHARGING MODES OF ELECTRIC VEHICLES

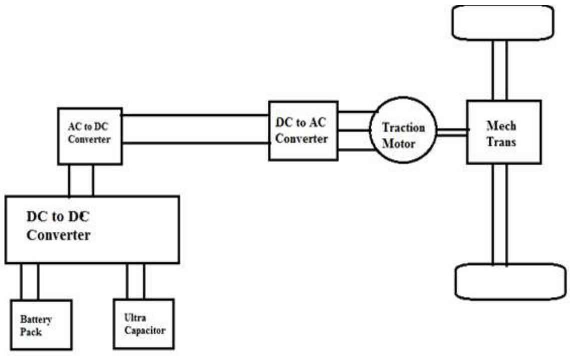

Fig. 5 Electric Vehicle Architecture

Fig. 5 shows an electric car's architecture, which incorporates inverters and converters. Depending on the requirements, the batteries' output voltages are boosted or bucked and converted to AC. as the motor needs an AC input and the battery is a DC supply. DC power is provided to the transformer's main winding within the casing of the inverter by an electronic switch, and the direction of the current flow is periodically changed. As a result, the transformer's secondary winding circuit produces AC current [9]-[12].

For EVs, there are now realistic options for fast charging, public charging, and private home charging. The Electric Vehicle Service Equipment (EVSE) acts as a conduit between the charging port and the power supply to safely provide AC energy to the vehicle. Depending on how rapidly the battery is charged, multiple charging rates are available [6].

A.Level-1 Charging

A plug must be used to connect to the onboard charger when charging at level 1.This type of charging is the most time-consuming and is often used in homes. It is powered by 120V AC and has current ratings either 15Aor 20A. Between eight and sixteen hours may be required to completely charge an EV battery.

B. Level-2 Charging

Because they operate at 220 V or 240 V, Level 2 chargers charge faster than Level 1. In order to install a charging device, a cable, and a power source, a 240 V AC power supply with a 40 A current rating is needed for private premises, while a 400 V AC power source with an 80 Afor public installations, it is necessary to have a current handling capacity. In 4 to 8 hours, an EV battery may be completely charged.

C. Level -3 Charging

Level-3 charging, sometimes referred to as DC fast charging, creates a high-power DC current of 120 kW and links it directly to the battery using an off-board charger. DC fast chargers have been installed in public and business areas. This charging technique offers an 80% charge in 10–15 minutes. It is one of the most important strategies and it will significantly affect how electric cars are charged in public. It needs three-phase circuits with a 480 V or higher voltage rating.

IV. ENERGY STORAGE SYSTEM

The high-voltage batteries, which make up the majority of the car's electric system, should be positioned to have a low center of gravity and improve stability. Currently, electrochemical battery packs are the most popular kind of energy storage for electric cars [19]. Battery management systems (BMS) keep track of, control, and safeguard electric vehicle batteries from injury. Changes in the impedance, temperature, and self-discharge properties of the cell cause variations in the crucial BMS function of cell balancing [20]. The battery pack's passive mechanism uses balancing resistors to drain surplus charge from overcharged cells in order to balance the cell voltages evenly. Electrical energy is stored using batteries and other electrochemical devices. Batteries are divided into three classes based on the size of the unit: cells, modules, and battery packs. [22]-[23].

A.Functions of Battery Management Systems

1) *Charging Control*

An electric vehicle's battery is charged twice utilizing constant voltage and current. Throughout the first level, the charger maintains a steady current, raising the battery's voltage. The battery achieves the constant voltage stage when the voltage reaches a constant number, which happens when the battery is almost full. The battery current decays exponentially whereas the voltage remains constant until the charge is completed.

2) *Discharging* Control

When a cell is emptying, the battery may be vulnerable to under-current and under-voltage situations. It is possible for overvoltage and over current conditions to occur when the battery is charging, which would quickly increase the battery's internal temperature. The BMS's role is to protect the cell from any occurrences that could cause it to exceed its limitations.

3) *State-of Charge Determination*

The SOC is responsible for controlling the cell charging and discharging as well as sending user signals. Direct measurement, coulomb counting, or a mix of the two methods can be used to determine SOC [24]-[25].

4) *State-of Health Determination*

SOH measures battery's overall health and is a gauge of its long-term capacity. It also indicates the performance of specific parameters when compared to a new battery as a reference. To determine the cell's SOH, any characteristics that change considerably with age, such as cell impedance or cell conductance, could be employed. It could be determined using a single-cell characteristic, such as cell impedance or conductance [26]-[27].

5) *Cell Balancing*

Electric motors and their accessories are powered by series-connected battery cells. These cells behave differently depending on the situation, compared to how a battery charges and discharges.Each cell can have a different voltage and current than the others, which could lead to overcharging or undercharging in some cells. These could lead to internal short circuits and early cell deterioration as a result of the separator, cathode, and anode of the battery being deformed. Cell balancing is used to control both energy distribution in EVs and voltage levels of the cells in order to address this issue by reducing the battery for electric vehicles' power consumption [28].

Even though they are made of the same material or by the same manufacturer, no two cells are the same. State-of-charge, discharging time, impedance, temperature, and capacity are all slightly different from one another. Variations in voltage over time may be caused by changes in a single cell's characteristics. The charger normally checks the voltage throughout the whole string of cells to determine if it has reached the voltage regulation point or not. [31]-[32].

Run-time efficiency and battery life are both impacted. Two methods such as passive and active can be used to maintain cell balance. The batteries that must be balanced are discharged using a resistor-based dissipative bypass route. Another name for passive battery charging isresistor bleeding balancing. Passive cell stability is a sort of it. Energy is directed to where it is required rather than being squandered using the active cell balancing approaches, although it is more costly [33].

B. *Battery Management Systems Topology*

The three most common topologies for BMS hardware design are discussed as follows [29]-[30]:

1) *Distributed Topology*

Fig. 6 shows the distributed topology of BMS, each cell in this topology has a BMS board which is equipped with a single digital communication signal that may be used to switch off or separate the charger as well as report its status. Although it demands a huge number of PCBs, it is simple and dependable (printed circuit board).

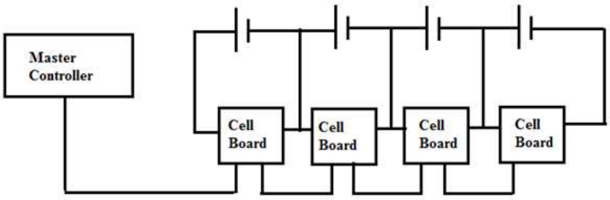

Fig. 6 Distributed Topology of BMS

2) *Modular Topology*

In this design, each cell has a BMS board which is equipped with a single digital communication signal that may be used to switch off or separate the charger as well as report its status although it demands a huge number of PCBs as shown in Fig.7, it is simple and dependable (printed circuit board) [18].

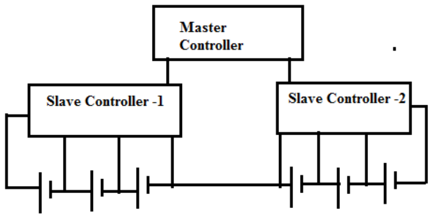

Fig. 7 Modular Topology of BMS

3) *Centralized Topology*

Each battery cell has a connection that connects it directly to the master control unit as in Fig. 8. It is a single controller. This architecture allows for simple inter-vehicle communication with just a few wires, but it generates lot of heat since the controller is necessary for central positioning as it is the main source of cell balancing in an EV (cells are dispersed).

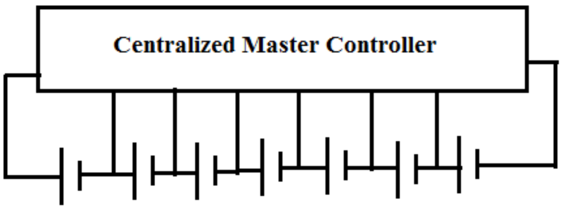

Fig. 8 Centralized Topology of BMS

C. *Battery Thermal Management*

Temperature is impacted by operational situations, battery health, structural design, and thermal management, which is significantly influence the battery performance and health. The temperature has a substantial impact on the efficiency of a lithium-ion battery, and thermal management systems must account for heat production in cells, temperature change among cells, and heat transfer with the environment. Because the cell's internal resistance rises at low temperatures, heating is needed when the temperature is low.Thus, cooling is essential for functioning of the cells throughout their lifespan. The three various temperature-related methods include 1) maintaining a constant ambient temperature while disregarding battery temperature; 2) testing at various ambient temperatures, and 3) employing easily available data for a cell and/or ambient temperature [31].

V. MACHINE LEARNING APPROACHES FOR BMS

At first, the concepts of AI and ML as well as problems with battery management in future electric cars were the main topics. In order to optimize processing power and battery data availability, machine learning algorithms are among the most efficient ways to identify important battery performance indicators, such as SOH, SOC, and the battery's remaining usable life. Better battery monitoring approaches are necessary as EV usage grows and greater voltage, higher efficiency, and longer lifetime battery systems are needed [32]-[33].

Similar to a fuel gauge found in the classic gas-powered vehicle, the battery SOC gauges how much useful energy is still there in the battery. Numerous techniques and processes are used to do this, and they all rely on signals that can be observed, such as the voltage, current, and temperature of the battery terminals. The nonlinearity of the battery makes this a challenging task. As a consequence, there is a high correlation between battery capacity and the assessment of various charging patterns (CPs). Some of these characteristics contain temperature, voltage, and current, and as batteries evolve and undergo more cycles, the patterns of the CPs change. In conclusion, while calculating a battery's capacity [34,machine learning (ML), a rapidly expanding field, is used in a wide variety of ways. Machine learning techniques could be categorized into three fundamental categories: supervised learning (A), unsupervised learning (B), and reinforcement learning (C). It is crucial to develop a versatile, self-reconfigurable, and trustworthy model in order to accurately monitor and regulate the battery's state. [35]-[36].

A variety of SOC estimating techniques has been created recently, with model-based approaches, data-driven techniques, look-up tables, and coulomb counts being a few examples [37].

Compared to model-based approaches used in coulomb counts, look-up tables, and related techniques, data-driven strategies have a number of benefits. Data-driven methodologies employ machine learning technology to derive correlations and rules from data. Data-driven approaches are more precise, less expensive, and quicker to build without requiring in-depth subject knowledge. Support vector machines, fuzzy logic systems, artificial neural networks, deep neural networks, and other data-driven methodologies are some of the alternatives [38].

Some of the ML based SOC estimation methods are Extreme Learning Machines (ELM), Fuzzy Logic Systems, Artificial Neural Networks, and Vector Machines. These approaches are contrasted using a variety of factors, including computational costs, accuracy, storage error, and many more. Data-driven techniques for detecting SOC and SOH, such machine learning are becoming more and more popular as battery data becomes more readily available and processing capacity increases[39]-[42]. The following methods are applicable for BMS by using ML approaches.7

A. *Support Vector Machines*

A novel approach that applies kernel functions to a variety of learning problems must be developed in light of recent advancements in Support Vector Machine (SVM) [43].

B. *Fuzzy Logic Systems*

When a suitable database is provided, fuzzy logic is a further useful technique for characterizing the non-linear behavior of the battery. A typical fuzzy system often replaces complex mathematical statements with a set of fuzzy rules. The three stages of a fuzzy logic estimator are fuzzy classification, rule-based inference, and defuzzification. The input values are now mapped to the relevant fuzzy sets. The membership level, sometimes referred to as the truth value, is subsequently applied to these mapped values [44].

C. *Artificial Neural Network*

Artificial neural networks (ANNs) are constructed based on the principles of how the human brain interprets data. At its most fundamental level, an ANN is composed of three layers: an input layer, a hidden layer, and an output layer. The "neurons" that link each of these levels to the others are each connected by a specific set of weights. Back propagation neural network is the technique used by ANNs to train their networks (BPNN). It facilitates weight calculations [45].

D. *Deep Neural Network*

The complicated behavior of the data is better understood by deep neural networks, which use a hierarchical concept structure. By analyzing its non-linear behavior, the battery may be made ideal for preventing the long-term error accumulation effect.

The two different types of deep neural network-based algorithms are recursive algorithms and convolution algorithms. A recurringneural network is the best technology to tackle time series problems, utilizing the right weight values to address complex systems, and their special dynamic memory. The three modules that make up the convolution neural network model are the training, testing, and verification modules. [46].

VI. CONCLUSION

Electric vehicles, their impacts, the numerous charging techniques, and the battery management system are all included in this review research. As the world works to reduce carbon pollution and oil dependency, both of which are harmful to the environment and raise greenhouse gas emissions. Electrification of vehicles has gained relevance since it will also reduce other harmful pollutants such as ozone and particle matter. The need for electrical energy is increasing as the number of EV's grows drastically. In future, EVs will be given more importance because of the depletion in the battery costs. By collecting information from the experiment and the internet environment, machine learning—a relatively new technique—can be used to forecast battery life. This idea of machine learning is used in data-driven ways to accurately estimate and predict the SOC of batteries. A data-driven modeling will be a useful method for battery prognostics and diagnostics. It also encourages the development, improvement, and launch of new applications.

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